

MANDATORY SUMMARY TABLE

Decade-Aware Compliance Assessment Matrix

Aspect	Covered in Current Paper	Adequate Issues for 2025	Issues Identified	Recommended Action
1. TEMPORAL FOCUS (2015-2025)	✓ (80%+)	✗	Cutoff ~2022; post-2022 paradigm shifts severely underrepresented (transformers, LLMs, foundation models)	Add Section 6: "Transformer & Foundation Model Era (2023-2025)"; extend temporal reach to Dec 2024 minimum
2. STATIC ANALYSIS COVERAGE	✓	✓	PE headers, n-grams, opcodes comprehensively covered; obfuscation limitations noted	Keep existing; add note on IR-based semantics as modern alternative to syntactic analysis
3. DYNAMIC ANALYSIS COVERAGE	✓	✓	Cuckoo, DBI, anti-evasion techniques well-detailed (Tables 2, 3, 6)	Keep existing; add DRAKVUF, lightweight containerization (SecBox); update anti-evasion taxonomy post-2023
4. GRAPH NEURAL NETWORKS (GNN)	✗ (minimal)	✗	Only CFGExplainer [4] mentioned in passing; GNN paradigm for malware analysis ignored	Create Section 5 subsection "Graph-Based Structural Analysis (2021-2023)"; cite [web:4, 7, 10, 35]; discuss GCN/GAT/GIN on CFG
5. BINARY LIFTING & IR	✗	✗	Completely absent; critical for semantic robustness to obfuscation	Create Section 5 subsection "Binary Lifting & Intermediate Representations (2017-2025)"; cite [web:3, 14, 17, 20]; explain Valgrin/BAP/LLVM IR approach

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6. TRANSFORMER ARCHITECTURES	x	x	No dedicated section; only implicit mention in LLM context; transformer paradigm not centered despite SOTA status Create Section 6 subsection "Transformer-Based Malware Analysis (2023–2025)"; cite [web:5, 8, 11]; cover BinBert, Vision Transformers, assembly transformers
7. LLM-ASSISTED ANALYSIS	x (4 papers only)	x	Treated as sidebar to generative AI; actually represents paradigm shift from feature engineering to semantic understanding Create major Section 6 "LLM-Assisted & Semantic Analysis (2023–2025)"; 5+ pages; cite [web:24, 27, 28, 40]; discuss CodeLLaMA, GPT-4, semantic lifting
8. FEW-SHOT & META-LEARNING	x	x	Completely absent; critical for novel/zero-day malware detection Add to Section 6; cite [web:23, 29]; discuss SIMPLE, graph-based meta-learning
9. TEST-TIME ADAPTATION	x	x	Not mentioned; important for handling concept drift Add to Section 6; cite [web:35, 38]; discuss MADCAT, warm-start learning, pseudo-labeling
10. ADVERSARIAL ROBUSTNESS	x (evasion only)	x	Anti-analysis techniques covered extensively; robustness of detectors against adversarial malware is minimal Expand into new Section 7.6 "Adversarial Robustness & Robustness Testing"; cite [web:13, 19, 35]; cover GAN evasion, certified defenses
11. EXPLAINABILITY (XAI)	x	x	SHAP, LIME, Grad-CAM not mentioned; Add to Section 7 "Interpretability & Explainability (2019–2025)"; cite

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			critical for trust & debugging	[web:22, 25, 31]; discuss feature importance, decision explanation
12. CONCEPT DRIFT & TEMPORAL	✗ (mentioned challenge)	✗	Acknowledged as problem; no evaluation framework provided ; time-stratified train/test absent	Add Section 7.2 "Temporal Evaluation & Concept Drift"; cite [web:32, 35, 38]; recommend time-split evaluation methodology
13. CROSS-DATASET GENERALIZATION	✗	✗	Generalization failures noted; no systematic evaluation protocol ; train on EMBER, test on VirusShare not standard	Add Section 7.3 "Cross-Domain & Cross-Dataset Robustness"; cite [web:33]; protocol: train on A, test on B, C, D
14. DATASET: EMBER	✓	✗	Acknowledged; saturation issue (0.999 AUC baseline) NOT discussed	Clarify: EMBER is saturated benchmark; note that 98-99% accuracies may reflect dataset ease, not detector capability
15. DATASET: SOREL-20M	✓	✓	Good coverage; 20M scale noted	Keep; emphasize time-split recommendations for temporal evaluation
16. DATASET: BODMAS	✓	✓	Timestamp-aware; good for temporal analysis	Keep; note advantages over EMBER for temporal studies
17. DATASET: EMBER2024	✗	✗	COMPLETELY MISSING (released June 2025); 3.2M samples, 7 tasks, evasion-focused challenge set	Critical gap: Add EMBER2024 to Section 4.3; major new section header; emphasize evasion-focused evaluation
18. EVAL: TEMPORAL / DRIFT	✗	✗	Concept drift acknowledged;	Add Section 7.2: Define concept

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			no temporal evaluation framework provided; no discussion of how models degrade over time	drift formally; recommend time-stratified splits; cite temporal drift empirical studies
19. EVAL: CROSS-DATASET	x	x	Generalization failures discussed; no standard cross-dataset protocol ; most papers train/test on same dataset	Add Section 7.3: Propose protocol (train EMBER, test on SOREL/ VirusShare/ BODMAS); measure domain gap
20. EVAL: ADVERSARIAL ROBUSTNESS	x	x	No systematic evaluation of detector robustness to adversarial/ evasive samples	Add Section 7.6: Evaluate against GAN-based evasion, certified defenses; cite adversarial robustness literature
21. LEARNING PARADIGM: Classical ML	✓	✓ (baseline)	Covered adequately	Reposition: "Baseline Approach (Lightweight, Interpretable)"; clarify when RF/ SVM preferable to DL
22. LEARNING PARADIGM: Early DL	✓	x (over-emphasis)	CNN/RNN vanilla treated equal to modern architectures; not clearly marked as transitional/ superseded	Demote to subsection "Early Deep Learning Attempts (2015–2018): Foundations & Limitations"; cite generalization failures
23. LEARNING PARADIGM: Hybrid Static+Dynamic	✓	x	Good examples (AI-Hydra, CSPE-R); not framed as pragmatic engineering choice	Reframe as "Hybrid Fusion Approaches (2018–2021): Engineering Trade-Offs"; discuss accuracy vs. latency vs. cost
24. LEARNING PARADIGM:	x	x	Self-supervised pre-training,	New subsection in Section 5:

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Representation Learning			contrastive learning, embeddings absent	"Representation Learning & Self-Supervised Pre-Training (2020-2023)"; discuss why paradigm shifted
25. LEARNING PARADIGM: Graph-Based	x	x	GNN paradigm for malware analysis not positioned as SOTA for structural data	New subsection in Section 5: "Graph Neural Networks on Control/Data Flow (2019-2023)"; discuss GCN, GAT, GIN
26. LEARNING PARADIGM: Transformers	x	x	Not dedicated subsection; transformer SOTA status not established	Major subsection in Section 6: "Transformers for Malware Analysis (2023-2025)"; cover BinBert, Vision Transformers, assembly transformers
27. LEARNING PARADIGM: Foundation Models	x	x	LLM mention buried; paradigm-shifting importance underappreciated	Major subsection in Section 6: "Foundation Models & Large Language Models (2023-2025)"; 5+ pages; discuss transfer learning, fine-tuning
28. PARADIGM CLARITY TABLE	x	x	Figure 4 presents ML vs. DL as binary; conflates distinct paradigms; no "what should readers use in 2025?" guidance	Replace Figure 4 with revised "Learning Paradigms: Recommendation for 2025" table; stratify: legacy
29. OBFUSCATION: Packing & Polymorphism	✓	✓	Baseline obfuscation covered	Keep; remains foundational
	✓	✓	Comprehensive taxonomy (Tables	Keep; mature coverage

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30. OBFUSCATION: Anti-Analysis (Debugger/VM/DBI)			2-3, 6) with empirical prevalence data	
31. OBFUSCATION: Semantic & CFG Flattening	x	x	Control-flow flattening, opaque predicates not discussed; emerging technique not covered	Add Section 5: "Semantic Obfuscation & Control-Flow Transformations (2021-2025)"; discuss GNN robustness to flattening
32. OBFUSCATION: AI-Powered Evasion (GANs)	✓ (brief)	x	DeepLocker, GUI attacks mentioned; GAN-based feature evasion & adversarial malware sparse	Expand Section 7.6: Comprehensive GAN taxonomy; cite [web:13] for 2025 overview of adversarial evasion
33. OBFUSCATION: Evolution Narrative	x	x	Obfuscation techniques listed; no "why did this evolve in response to detection" narrative	Add timeline figure showing obfuscation evolution decade-wise; explain pressure from improved detection
34. PRACTICALITY: Scalability	✓ (implicit)	x	Mentions DBI slow; no formal analysis of real-world volumes (450K samples/day)	Add Section 7.1 "Scalability & Throughput Requirements"; discuss 450K/day reality; computational bottlenecks
35. PRACTICALITY: Latency & Resource Cost	✓ (mentioned)	x	AI-Hydra latency (60.9 sec) noted; no systematic comparison table across models/methods	Add to Section 7.1: Table with model, inference latency, memory, GPU requirement, FPS achieved
36. PRACTICALITY: Tiered Detection Pipelines	x	x	Hybrid static+dynamic discussed; not framed as production pipeline (fast → deep → manual)	Add Section 7.1: Design pattern with cost analysis; recommend tiered approach

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37. PRACTICALITY: Tooling & Infrastructure	✓ (named)	✗	Tools listed (IDA, Cuckoo, Pin); no discussion of reproducibility, versioning, containerization	Add Section 11 "Tools, Infrastructure & Reproducibility"; recommend Ghidra (free, open-source); discuss Docker; version pinning
38. PRACTICALITY: Anti-Analysis Arms Race	✓	✗	Extensively covered; doesn't question viability of static analysis given 60–80% obfuscation prevalence	Add commentary in Section 7: Is static analysis becoming obsolete? Discuss IR-based & semantic alternatives
39. PRACTICALITY: Cascading/Staged Pipelines	✗	✗	Hybrid mentioned; not framed as realistic production cascade	Add Section 7.1: Tier 1 (static, <1s) → Tier 2 (dynamic, 30–60s if uncertain) → Tier 3 (manual, critical) Add Section 11: Recommend papers release code
40. REPRODUCIBILITY: Code & Data Availability	✗	✗	Not discussed; papers analyzed without noting open-source availability	(GitHub), datasets (with legal path), Dockerfiles; discuss reproducibility crisis
41. TAXONOMY: Temporal Organization	✗	✗	Current structure (functional: analysis → features → methods) lacks temporal progression	Restructure entire paper using decade-layered approach (proposed Section 8); show evolution not just snapshot
42. TAXONOMY: Visual Figure	✗	✗	No evolution timeline figure; no maturity matrix; Figure 4 conflates paradigms	Create Figure 1: Decade-wise technique evolution; Figure 2: Maturity matrix (technique × era × maturity level)
	✗	✗	Paper doesn't clearly say "use	Add Table: "Learning

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43. TAXONOMY: Clarity for Practitioners			this method in 2025" vs. "this is historical"	Paradigm Recommendations for 2025"; map technique → legacy
44. STRUCTURE: Logical Flow	✓	✓	Current organization (analysis → features → data → methods) makes sense	Keep organization but add temporal layering within sections
45. STRUCTURE: From-Scratch Guide	✗	✗	No guidance for grad students writing similar survey; methodology section brief	Add Section 9: "Systematic Review Strategy (Best Method for 50-100 Papers)"; discuss LLM-assisted workflow
46. WRITING QUALITY: Clarity	✓	✓	Paper well-written, figures clear, tables comprehensive	Keep; minor proofreading pass
47. WRITING QUALITY: Completeness	✓	✗	Comprehensive within scope; major topic gaps (GNN, transformers, LLMs, robustness, drift)	Address via new sections
48. CITATION: Breadth (2015-2025)	✓	✗	77 papers included; range spans decade; distribution skewed toward 2018-2022 (>60% of papers)	Rebalance: aim for 30% pre-2020, 40% 2020-2022, 30% 2023-2025
49. CITATION: Pre-2015 Papers	✓ (foundational)	✗ (over-cited)	Foundational papers (e.g., Ye 2017, Gibert 2015) cited throughout main narrative	Demote to background/ footnotes; emphasize pre-2015 as historical not current
50. CITATION: Post-2022 Papers	✗	✗	Minimal post-2022 coverage; LLM papers sparse;	Add 15-20 post-2022 papers; cite [web:3, 5, 8, 15, 24, 27, 28, 31, 32, 35, 38, 40, 41]

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51. CITATION: High-Impact Omissions	x	x	paradigm-shift papers missing BinBert, CFGExplainer, Security LLM, EMBER2024, SoK papers, concept drift papers absent	Systematic addition of [web:3, 4, 8, 15, 24, 27, 28, 31, 32, 35, 38, 40, 41]
52. READING LIST	x	x	No prioritized guide for readers; no stratification by era/difficulty	Add Appendix: Stratified reading list (foundations → core → modern → emerging); ~30-40 papers
53. OPEN PROBLEMS: Identified	✓	x	Challenges mentioned; not systematically listed or prioritized	Add Section 9: Explicit open problems list; prioritize by impact + feasibility
54. FUTURE DIRECTIONS: Established vs. Speculative	x	x	Conclusion mentions future work; no distinction between proven trends and speculative	Add Section 9.2: Separate "Established (high confidence)" from "Emerging (medium)" from "Speculative (low)"
55. FUTURE DIRECTIONS: Actionable	x	x	No concrete research roadmap; grad students can't use to decide next project	Add Section 9.3: Prioritized recommendations (high impact + feasible first)
56. OVERALL NARRATIVE	✓	x	Coherent within 2018-2022 scope; lacks decade-aware progression showing why techniques evolved	Restructure to show: Legacy (pre-2015) → Classical (2015-2018) → DL Emergence (2018-2021) → Structural (2021-2023) → Transformers (2023-2025)
57. CONTRIBUTION POSITIONING	✓	x	Paper claims to cover "state-of-the-art methods"; actually covers	Retitle: "Static & Behavioral Malware Analysis with Machine

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58. SUBMISSION READINESS	✓	✗	2015-2022 SOTA; 2023-2025 paradigm shifts missed	Learning (2015-2025): Evolution from Feature Engineering to Foundation Models"
			Paper structure coherent, publishable in 2023; insufficient for 2025 standards without major revision	Estimate: 4-6 weeks for moderate revision; 8-12 weeks for comprehensive rewrite

SCORING SUMMARY

Category	Current Coverage	2025 Adequacy	Severity	Effort to Fix
Temporal Scope	60%	40%	CRITICAL	High (restructure)
Technique Breadth	70%	30%	CRITICAL	High (major additions)
Dataset Discussion	80%	50%	HIGH	Medium (update table)
Learning Paradigm Clarity	50%	20%	CRITICAL	High (reframe)
Practical Deployment	40%	30%	HIGH	Medium (new section)
Robustness & Evaluation	30%	20%	HIGH	High (new sections)
Literature Quality	75%	45%	CRITICAL	Medium (add citations)
Taxonomy & Structure	65%	35%	HIGH	High (restructure)
Writing Quality	85%	80%	LOW	Low (proofreading)
Reproducibility	20%	10%	MEDIUM	Low (add section)

OVERALL RECOMMENDATION

For Current Paper Revision:

- **Quick (1 month):** Add 2–3 pages on transformers/LLMs; update datasets to EMBER2024; clarify temporal scope in title
- **Moderate (4–6 weeks):** Add sections on GNN, binary lifting, robustness, practical deployment; revise learning paradigm section
- **Comprehensive (8–12 weeks):** Full restructure into decade-layered organization; complete paradigm narrative; 10+ new pages

For From-Scratch Rewrite:

- **Timeline:** 6–9 months (30-week structured workflow with LLM assistance)
- **LLM Role:** ~40% (summarization, extraction, scaffolding); always human-validated
- **Output Quality:** Discipline-shaping survey with clear decade-first perspective + actionable research roadmap

Submission Readiness:

- **Current:** Publishable as-is in 2023; insufficient for 2025 venue standards
- **After revision:** Competitive for top venues (IEEE S&P, USENIX Security, ACM Computing Surveys) if major content added
- **Key:** EMBER2024 addition is non-negotiable for 2025 credibility

This table provides quick-reference assessment for authors, reviewers, and survey developers. For detailed reasoning, see main audit report (1,500+ lines of analysis).